■■ CallShield: Master Technical Specification & Implementation Guide

A Comprehensive Real-Time Acoustic Forensic & Telephony-Integrated Anti-Vishing Platform

Project Lead & Developer: Kalyan Kumar	**Core Classifier:** ExtraTrees Ensemble (F1 = 0.9038)
Client Framework: React Native & Expo SDK 52 (Android)	**Feature Vector:** 60 Physical & Acoustic Biomarkers
Backend Engine: FastAPI + Uvicorn (Render Cloud)	**Inference Speed:** < 15 ms per 2.5s Audio Chunk
Database: Supabase PostgreSQL + AsyncStorage	**Telecom Filter:** GSM 300Hz-3400Hz Bandpass + 18dB SNR

1. Executive Overview & Vishing Threat Landscape

The Cybersecurity Threat: Over the past two years, the emergence of generative neural text-to-speech (TTS), latent diffusion vocoders, and zero-shot voice cloning tools (such as ElevenLabs, WaveNet, and Voice-to-Voice conversions) has weaponized voice synthesis. Cybercriminals require only 3–5 seconds of a victim's reference audio to clone their vocal timbre, prosody, and emotional inflections with indistinguishable realism. This has catalyzed a massive surge in **vishing (voice phishing)** crimes, including:

- **Bank OTP & Credential Harvesting:** Cloned IVR bots and fake financial officers deceiving victims into disclosing multi-factor authentication codes.
- **Emergency Kidnap & Extortion:** Impersonating family members in distress to demand immediate ransom transfers.
- **Executive CEO & Wire Fraud:** Replicating corporate leadership voices to mandate unauthorized corporate wire transfers.

Why Current Solutions Fail: Conventional caller ID applications (e.g., Truecaller) operate strictly on static databases of reported telephone numbers. Because attackers utilize Voice-over-IP (VoIP) caller ID spoofing and disposable virtual SIMs, number blacklists are bypassed instantly. *Existing defenses perform zero acoustic inspection of the live voice.*

The CallShield Solution: CallShield is a real-time mobile and cloud security platform that intercepts cellular phone calls natively on Android, streams 2.5-second live audio chunks, extracts 60 physical vocal biomarkers (micro-jitter, shimmer, ΔΔ-MFCCs, spectral flatness), and evaluates voice authenticity using an ExtraTrees Ensemble in under 15 milliseconds. When an AI clone is detected, a full-screen red siren is triggered with automated 1-tap call termination.

2. System Architecture & Component Interactions

CallShield utilizes a multi-tier collaborative edge-cloud architecture designed for microsecond latency, zero data retention, and continuous real-time telemetry:

System Layer	Technologies	Exact Role & Functionality
1. Telephony Client	React Native (Expo 52) Android Java Native Modules AsyncStorage	Subscribes to PhoneStateListener, intercepts CALL_STATE_RINGING, records 16kHz mono PCM, displays HUD overlays, and executes TelecomManager.endCall().
2. REST & WebSocket API	FastAPI (Python 3.11) Uvicorn / Gunicorn Render Container Cloud	Non-blocking asynchronous server managing streaming endpoints (/api/v1/predict/live), WebSocket telemetry, silence threshold gating, and threadpool scheduling.
3. DSP & Feature Engine	Librosa 0.10+ SciPy Signal SoundFile C-Bindings	Performs sliding window chunking (2.5s duration, 1.5s hop), peak amplitude normalization, and extracts 60 physical vocal biomarkers.
4. ML Classification	Scikit-Learn ExtraTrees VotingClassifier Attribution Engine	Evaluates 100 randomized decision trees to compute calibrated threat probabilities and attributes specific AI synthesis architecture (ElevenLabs, WaveNet, V2V).
5. Persistence & Cloud DB	Supabase (PostgreSQL) PBKDF2 Cryptography	Stores encrypted user authentication records, forensic chunk timelines (prediction_history), and crowdsourced scam blacklists (scam_blacklist).

3. Acoustic Signal Processing & The 60 Forensic Biomarkers

Audio signals are sampled at $f_s = 16\text{ kHz}$ mono. Continuous speech is segmented into sliding windows of $T = 2.5\text{ seconds}$ (40,000 samples) with an overlap hop of 1.5 seconds. Each chunk is transformed into a 60-dimensional feature vector $\mathbf{x} \in \mathbb{R}^{60}$:

Feature Group	Count	Mathematical Description & Biometric Significance
1. Static MFCCs	26	13 MFCC means + 13 MFCC stds. Captures the physical resonant cavity shape of the human vocal tract (formants) via Discrete Cosine Transform on Mel-scaled filterbanks.
2. Dynamic Δ-MFCCs	26	13 First-Order Delta means + 13 Delta stds. Quantifies the instantaneous velocity of spectral transitions across speech frames, exposing unnatural vocoder transitions.
3. Vocal Micro-Jitter	1	Measures cycle-to-cycle fundamental pitch period perturbation. Organic vocal cords exhibit natural micro-tremor ($1.0\% - 2.5\%$); AI models produce unnaturally smooth pitch periods ($< 0.1\%$).
4. Acoustic Shimmer	1	Cycle-to-cycle peak amplitude perturbation across glottal cycles. Evaluates sub-glottal pressure stability.
5. Spectral Flatness & Rolloff	4	Spectral flatness measures tonal-to-noise ratio (detects neural vocoder high-frequency artifacts). Spectral rolloff (R_{85}) pinpoints artificial frequency cutoff caps at 16kHz or 20kHz.
6. Spectral Centroid & ZCR	2	Spectral center of mass and Zero-Crossing Rate (ZCR) to measure unvoiced fricative energy and harmonic distribution.

4. Telecom-Grade Channel Modeling (Overcoming the Studio Trap)

The Studio Trap Problem: Machine learning models trained exclusively on full-spectrum studio audio ($20\text{ Hz} - 20,000\text{ Hz}$) fail catastrophically on real cellular calls because cellular telephone networks (GSM, VoLTE, PSTN G.711) truncate high and low frequencies. To ensure robust real-world performance, CallShield applies telecom-grade channel transformations:

1) **4th-Order Butterworth Bandpass Filter ($300\text{ Hz} - 3400\text{ Hz}$):** Filters out sub-bass and extreme treble, matching native GSM cellular network codecs.
2) **Gaussian Channel Noise Injection (18 dB SNR):** Simulates ambient room acoustics and microphone static.
3) **Downsampling to 16 kHz Mono PCM:** Matches mobile telephony hardware capture pipelines.

5. Machine Learning Architecture: ExtraTrees Ensemble

Model Selected: **ExtraTrees (Extremely Randomized Trees) Classifier** (`ml/models/baseline_model.joblib`) consisting of 100 randomized decision trees.

Mathematical Formulation: At each split node, ExtraTrees selects a random subset of $K = \sqrt{P} = \sqrt{60} \approx 8$ features. For each feature, a split threshold a_k^* is drawn *uniformly at random* from $[\min(x_k), \max(x_k)]$, and the threshold maximizing Gini Impurity reduction is chosen:

$$\Delta I(S, a^*) = I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R), \quad I(S) = 1 - \sum_{c \in \{0, 1\}} p_c^2$$

Variance Reduction & Sub-15ms Speed: By randomizing thresholds, tree correlation ρ drops significantly. Total ensemble variance $\text{Var}(\bar{f}) = \rho \sigma^2 + \frac{1-\rho}{M} \sigma^2$ is minimized without expensive threshold-sorting algorithms, enabling sub-15ms execution during live telephone calls.

6. Dataset Composition & Benchmark Performance

Dataset Composition (383 Total Audio Chunks):
• **178 AI-Generated Chunks:** Kaggle Deepfake Voice (`birdy654/deep-voice-deepfake-voice-recognition`), ElevenLabs Neural Voice Clone v2, Google WaveNet, gTTS (English + Hindi vishing scripts), OpenAI Voice, and V2V pitch conversions.
• **205 Authentic Human Chunks:** Spontaneous multi-speaker conversations across diverse genders and dialects.
• **Partitioning:** 294 Training Chunks (77%) vs. 89 Unseen Testing Chunks (23%) using Stratified Splitting.

Model Architecture	Accuracy	Precision (AI)	Recall (AI Scams)	F1-Score	Latency
■ **ExtraTrees (CallShield)**	**88.76%**	**82.46%**	**100.00% (47/47)**	**0.9038**	**< 15 ms**
Soft Voting Ensemble	84.27%	81.13%	91.49%	0.8600	~ 35 ms
Random Forest	83.15%	80.77%	89.36%	0.8485	~ 22 ms
Gradient Boosting	82.02%	80.39%	87.23%	0.8367	~ 18 ms

7. AI Generator Attribution & Forensic Fingerprinting

CallShield's attribution engine (`model_attribution_service.py`) maps acoustic spectral metrics to the exact synthesis engine:

- **ElevenLabs Neural Voice Clone v2:** Spectral centroid $> 2800\text{ Hz}$, pitch std $\sigma_p < 7.0\text{ Hz}$, cutoff cap at 20.0 kHz (Diffusion TTS, 96.8% confidence).
- **Google WaveNet / gTTS:** Low pitch std $\sigma_p < 4.0\text{ Hz}$, elevated $\text{ZCR} > 0.08$, cutoff cap at 16.0 kHz (Autoregressive/Concatenative TTS, 98.5% confidence).
- **Real-Time Voice Conversion (V2V):** High RMS energy > 0.04, $\sigma_p \ge 8.0\text{ Hz}$, phase discontinuity index 0.92 (Pitch-Shift Masking, 94.2% confidence).
- **OpenAI Voice / PlayHT:** Transformer spectral envelope, flatness 0.82, cutoff cap at 18.0 kHz (Audio Transformer, 95.0% confidence).

8. Complete API Specifications & Database Schemas

A. FastAPI Endpoints & Response Schema

Endpoint: POST /api/v1/predict & POST /api/v1/predict/live


```
{ "predicted_label": "ai_generated", "ai_probability": 0.8950, "ai_confidence_percent": 89.5,
"overall_risk_score": 89.5, "risk_level": "high", "warning": true, "attributed_generator": "ElevenLabs Neural
Voice Clone v2", "synthesis_technique": "Neural Latent Diffusion TTS", "spectral_cutoff_khz": 20.0,
"chunk_count": 4, "chunk_predictions": [ {"chunk_id": "chunk_001", "start_time_seconds": 0.0,
"end_time_seconds": 2.5, "predicted_label": "ai_generated", "ai_probability": 0.8840, "risk_level": "high"},
{"chunk_id": "chunk_002", "start_time_seconds": 2.5, "end_time_seconds": 5.0, "predicted_label":
"ai_generated", "ai_probability": 0.9120, "risk_level": "high"} ] }
```


B. Database Schemas (Supabase PostgreSQL)

- **users Table:** id (SERIAL), full_name (VARCHAR), email (VARCHAR UNIQUE), password_salt (VARCHAR), password_hash (VARCHAR), auth_token (VARCHAR UNIQUE), created_at, last_login_at
- **prediction_history Table:** id (SERIAL), filename (VARCHAR), predicted_label (VARCHAR), ai_probability (FLOAT), risk_level (VARCHAR), warning (BOOLEAN), chunk_count (INT), chunk_predictions_json (JSONB), created_at
- **scam_blacklist Table:** phone_number (VARCHAR PRIMARY KEY), report_count (INT), source (VARCHAR), created_at

9. Native Android Telephony Integration & Security

CallShield's native client application interacts directly with Android's core telephony stack (`CallInterceptorModule.java`):

1) **State Detection:** Android BroadcastReceiver listens to PHONE_STATE events and detects CALL_STATE_RINGING.
2) **Temporal Risk Formulation:** Multi-chunk call risk is aggregated dynamically: $R_{\text{call}} = 0.55 R_{\text{avg}} + 0.30 R_{\text{chunk}} + 0.15 R_{\text{max}}$.
3) **One-Tap Native Call Drop:** Upon High Risk verification ($R \ge 65\%$), the native module triggers TelecomManager.endCall() to drop the line.
4) **Zero-Retention Ephemeral Privacy:** In-memory Linux /tmp buffers are unlinked immediately post-prediction. No voice recordings are retained.

10. Evaluator & Conference Reviewer Q&A; Reference

Q1: Why ExtraTrees over deep neural networks (e.g., Wav2Vec 2.0 or ResNet)?

Deep transformers require 300MB–1GB RAM, GPU acceleration, and introduce 300–800ms latency. ExtraTrees executes in <15ms on standard mobile/cloud CPUs, requires only 2MB memory, achieved 100% Recall (47/47 AI scams caught), and resists acoustic overfitting due to randomized threshold splits.

Q2: How do you handle ambient room noise and cellular compression?

Our dataset pipeline applies telecom-grade channel augmentation (300Hz–3400Hz 4th-order Butterworth bandpass filter and 18dB SNR noise injection). Furthermore, an RMS energy gate (min_rms = 0.0001) skips silent pauses to prevent false alarms on background noise.

Q3: What happens if internet connectivity drops during an active call?

CallShield utilizes an offline-first architecture. Blacklists and sessions are cached locally in AsyncStorage. For offline inference, the model compiles into an on-device ONNX Runtime binary (<2MB) that runs locally on the phone's CPU with zero internet requirement.

Q4: How do you ensure user voice privacy?

Audio slices sent to the backend are processed as ephemeral temporary files in Linux /tmp and immediately unlinked (deleted) post-inference. Zero raw speech recordings are persisted on cloud servers.